

Department of Computer Science and Engineering

CSE3422: Software Engineering Lab

Project Report

Project Title: CityMind: Urban Planning and Decision-Support Platform

Team Name: Team Procastinators

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1. INTRODUCTION

CityMind is a project-based urban planning and decision-support web application designed to provide planners and developers with a structured digital workspace for planning and analyzing a selected development area. The system combines geographic information, infrastructure data, population-related information, environmental indicators, analytical calculations, future-demand prediction, planning recommendations, feasibility analysis, budgeting, and reporting into a single platform.

Instead of treating urban planning as a collection of separate activities, CityMind provides a connected workflow in which users can define a project area, visualize it through an interactive GIS map, divide the area into planning blocks, add roads and facilities, analyze existing conditions, identify infrastructure gaps, evaluate candidate locations, simulate future scenarios, and generate planning reports.

1.1 Motivation

Urban development requires decisions about roads, residential blocks, healthcare, education, green spaces, drainage, commercial areas, utilities, and other infrastructure. These decisions often depend on different types of information, including geographic data, population information, infrastructure conditions, environmental indicators, and future development requirements.

CityMind was developed to provide a unified digital workspace where a planner can define a selected development area, visualize its spatial structure, analyze existing facilities and infrastructure, identify service gaps, estimate future demand, and evaluate possible planning interventions.

Instead of treating maps, analytical data, recommendations, and planning scenarios as separate activities, CityMind connects them into a single project-based workflow. A user can select an area and gradually develop a structured plan consisting of blocks, roads, gates, schools, hospitals, parks, commercial areas, drainage systems, utilities, and other planning features.

The system is designed for early-stage urban planning and decision support. It does not replace professional planning judgment, statutory approval, or verified official datasets.

1.2 Objective

The primary objective of CityMind is to provide an integrated GIS-based planning platform that helps users analyze a selected development area, identify infrastructure and service gaps, evaluate suitable locations, understand future demand, compare planning scenarios, and create structured urban development plans.

The system combines GIS visualization, urban gap analysis, health assessment, growth prediction, risk detection, recommendation, location suitability, feasibility analysis, budget simulation, and reporting within a single project workspace.

1.3 Expected Outcome

The expected outcome is a responsive web-based urban planning platform where an authenticated user can create a planning project, define its boundary, divide the selected area into planning blocks, add infrastructure and facilities, perform analytical evaluations, receive location-based planning suggestions, simulate future scenarios, analyze budgets, and generate project reports.

The completed application provides a project-specific planning environment rather than a simple information or chatbot system. Users interact directly with maps, planning components, analytical dashboards, scenarios, and recommendations.

2. PROJECT DESCRIPTION

CityMind is a project-based urban planning and decision-support application designed for planning selected areas from their current condition toward future development.

A user can create a project for a new development, existing area, redevelopment, or urban expansion. After defining the project boundary, the system provides an interactive GIS environment where the selected area can be structured into blocks and populated with planning features such as residential zones, commercial areas, roads, gates, schools, hospitals, parks, drainage systems, utilities, and water bodies.

The overall CityMind workflow can be represented as:

Project & GIS Data → Gap Analysis → Urban Health → Growth Prediction → Recommendations
→ Location Suitability → Risk & Feasibility → Future Planning → Budget → Report

This workflow allows users to move from understanding the current condition of an area to developing and evaluating a structured future plan.

2.1 List of Features

The major functionalities of CityMind include:

1. User Registration, Login & Authentication
2. Private Planning Project Portfolio
3. Guided Planning Project Creation
4. Interactive GIS Planning Map
5. Block & Land-Use Planning
6. AI Urban Gap Analysis
7. Urban Health Score
8. Growth Prediction & Future Demand Analysis
9. Risk Detection
10. Smart Recommendation Engine
11. Exact-Location Suitability Analysis
12. Development Feasibility Analysis
13. Future Planning & Development Phases
14. Budget Simulation / Budget Optimizer
15. Report Generation
16. Project Validation & Project-Scoped Data Management

These features are integrated into a project workspace so that analytical results can be connected with the user's selected planning area.

2.2 Feature Details

2.2.1 User Authentication & Project Portfolio

CityMind provides user registration and login functionality. Users can securely access their own planning projects.

Authentication uses JSON Web Tokens (JWT), while passwords are protected using bcrypt hashing. Project ownership is enforced by the backend so that users can access and manage their own planning projects and associated map features.

2.2.2 Planning Project Creation

The application provides a guided multi-step project creation workflow.

Users can provide information including:

- Project name
- Project type
- Country and region
- Location
- Planning stage
- Population
- Number of households
- Population density
- Planning horizon

The system supports project types such as:

- New Development
- Existing Area
- Redevelopment
- Urban Expansion

After creation, the project becomes the central workspace for GIS planning and analysis.

2.2.3 Interactive GIS Planning

The GIS module is one of the central components of CityMind.

The system uses OpenStreetMap as the map data source and Leaflet/React Leaflet for interactive map visualization.

Users can:

- Define project boundaries
- Edit boundaries
- Import GeoJSON
- Export GeoJSON
- View project area
- Calculate boundary area

- Calculate project centroid
- Create planning blocks
- Add roads
- Add gates
- Define residential zones
- Define commercial zones
- Add schools
- Add hospitals
- Add parks
- Add utilities
- Add drainage
- Add water features

Planning features can also have different statuses such as proposed, recommended, approved, or rejected.

2.2.4 AI Urban Gap Analysis

The Urban Gap Analysis module identifies areas where the existing or planned infrastructure may not adequately support the expected population and development requirements.

The system evaluates areas such as:

- **Healthcare**
- **Education**
- **Parks**
- **Roads**
- **Commercial services**
- **Utilities**
- **Emergency services**
- **Drainage**

The system compares required capacity with existing, surrounding, and proposed capacity.

Users can also analyze current and future requirements over multiple planning horizons, including:

- **5 years**
- **10 years**
- **20 years**
- **30 years**

The system distinguishes between measured, estimated, and simulated information where applicable.

2.2.5 Urban Health Score

The Urban Health Score provides a structured measurement of the condition of a selected project or planning block.

The score considers factors including:

- Healthcare
- Education
- Mobility
- Environment
- Green space
- Infrastructure
- Accessibility

The backend calculates these scores deterministically, allowing the same inputs to produce reproducible results.

Users can compare current and planned conditions and perform what-if simulations involving facilities and infrastructure such as hospitals, schools, parks, roads, and drainage.

2.2.6 Growth Prediction

CityMind considers future development rather than focusing only on the current condition.

The Growth Prediction module estimates:

- Future population
- Future service demand
- Infrastructure requirements
- Healthcare demand
- Development requirements

The system supports multiple planning horizons so that planners can evaluate whether the proposed infrastructure can support future growth.

The prediction results are kept separate from measured data so that estimated future values are not presented as current facts.

2.2.7 Risk Detection

The Risk Detection module analyzes available project evidence and identifies potential planning risks.

Possible risk categories include issues related to:

- **Flooding**
- **Environmental conditions**
- **Traffic**
- **Infrastructure**
- **Accessibility**
- **Development constraints**

Depending on the available evidence, risks can be categorized into levels such as:

- **Low**
- **Moderate**
- **High**
- **Critical**

The purpose is to highlight areas requiring further investigation before development decisions are finalized.

2.2.8 Smart Recommendation Engine

The Smart Recommendation Engine provides interface-based planning recommendations rather than functioning as a conversational chatbot.

The system generates candidate interventions within the selected project area and evaluates them using deterministic planning logic.

Recommendation factors include:

- **Population need**
- **Infrastructure gaps**
- **Accessibility**
- **Future demand**
- **Existing service coverage**
- **Budget**
- **Planning horizon**

For example, if a selected block has insufficient healthcare coverage and sufficient accessibility, the system can identify the block as a candidate location for healthcare development.

Users can inspect recommendation candidates on the map and compare information such as estimated cost, population served, constraints, and supporting evidence.

2.2.9 Exact-Location Suitability Analysis

CityMind allows users to select an exact location on the map and evaluate whether a particular development could be suitable there.

The system evaluates factors including:

- Project boundary
- Land use
- Population need
- Service gaps
- Accessibility
- Existing facilities
- Future demand
- Planning constraints

This allows the user to move from a general recommendation to a specific location evaluation.

2.2.10 Development Feasibility

The Development Feasibility module provides a preliminary assessment of whether the proposed development plan is reasonably supported by the available project information.

It considers factors such as:

- Land area
- Population capacity
- Infrastructure gaps
- Urban health
- Risk information

The result is intended as a planning-readiness assessment rather than a replacement for engineering or statutory approval.

2.2.11 Future Planning

The Future Planning module allows a project to be divided into different development phases and scenarios.

For example, a project may be structured into:

Phase 1 → Roads, drainage and basic utilities

Phase 2 → Residential and commercial development

Phase 3 → Schools, healthcare and parks

Phase 4 → Future expansion and infrastructure upgrades

This allows planners to compare what should be developed immediately with what can be considered for later stages.

2.2.12 Budget Optimizer

The Budget Optimizer provides budget-based planning simulations.

The system supports different planning scenarios, including:

- **Minimum Cost**
- **Balanced**
- **Maximum Impact**
- **Baseline**

Users can adjust planning assumptions and evaluate how different development packages affect estimated cost and planning impact.

Budget simulation is integrated with project analysis so that financial planning can be considered alongside infrastructure and development requirements.

2.2.13 Report Generator

CityMind can assemble project information and analytical results into a structured report.

The report can include:

- **Project information**
- **Planning area**
- **GIS information**
- **Gap analysis**
- **Urban health results**
- **Growth prediction**
- **Risks**
- **Recommendations**
- **Feasibility**
- **Budget scenarios**
- **Future planning information**

Reports can be prepared for review, download, or printing.

3. TECHNOLOGY USED

3.1 Frontend

The frontend of CityMind is developed using React 19 with Vite.

The main technologies include:

- **React 19**
- **Vite**
- **Tailwind CSS**
- **React Leaflet**
- **Leaflet**
- **Recharts**
- **Axios**
- **Lucide React**

React provides the component-based application architecture, while Tailwind CSS and project-specific styling are used for the responsive user interface.

Leaflet and React Leaflet provide the interactive GIS environment, while Recharts is used for analytical visualizations and comparisons. Axios handles communication between the frontend and backend APIs.

3.2 Backend

The backend is developed using Node.js and Express.js.

It handles:

- **REST API endpoints**
- **Authentication**
- **Project management**
- **Validation**
- **Business logic**
- **Analytical services**
- **Recommendation calculations**
- **GIS-related operations**
- **Report-related operations**

The backend architecture is organized into routes, controllers, services, repositories, middleware, configuration, and utility modules.

Additional backend technologies include:

- **Zod for request validation**
- **JWT for authentication**
- **bcrypt for password hashing**
- **Helmet for HTTP security headers**
- **CORS for cross-origin access**

The analytical and recommendation calculations are primarily deterministic backend services so that outputs can be reproduced from the same inputs.

3.3 APIs

CityMind uses REST APIs to connect its frontend and backend.

The application provides API functionality for:

- **Authentication**
- **Cities**
- **Planning projects**
- **Planning features**
- **Gap analysis**
- **Block analysis**
- **Growth prediction**
- **Risk detection**
- **Reports**
- **Development feasibility**
- **Health simulations**
- **Validation**

- **Location evaluation**
- **Recommendations**
- **Future planning**
- **Budget simulation**

The frontend communicates with these services using Axios. OpenStreetMap provides the map basemap used by the GIS interface.

3.4 Database

CityMind uses MySQL 8 as its primary relational database.

The database contains structured information related to:

- **Users and roles**
- **Cities and wards**
- **Population**
- **Facilities**
- **Roads**
- **Environmental data**
- **Risk assessments**
- **Growth predictions**
- **Recommendations**
- **Impact simulations**
- **Planning projects**
- **Planning areas**
- **Planning features**
- **Project analyses**
- **Project scenarios**
- **Development phases**

Project boundaries and spatial feature geometries are stored in JSON/GeoJSON-compatible formats, while analytical attributes are stored using structured relational columns.

Foreign keys, indexes, unique constraints, status fields, and migrations are used to maintain data integrity and project-level ownership.

3.5 Testing

End-to-End Testing

Selenium is used to test browser-based workflows including:

- **Valid login**
- **Invalid login**
- **Project portfolio**
- **Project workspace**
- **GIS planning**
- **Growth prediction**
- **Gap analysis**

Manual Testing

Manual testing is also performed for:

- **Responsive UI**
- **Map interaction**
- **Project workflow**
- **Analytical dashboards**
- **Report generation**
- **Error handling**

The automated tests cover authentication, CORS, geometry, project context, urban scoring, location suitability, recommendation ranking, GIS interaction, project creation, workspace views, and report generation.

3.6 Git

Git is used for source-code version control throughout the development process.

The repository contains the major application components including:

- **Frontend**
- **Backend**
- **Database schema**
- **Migrations**
- **Tests**
- **Documentation**

Git allows the team to maintain development history, track changes, collaborate through branches, integrate completed work, and recover previous versions when necessary.

3.7 AI/ML Models

CityMind follows a hybrid AI/ML approach.

The core planning calculations are primarily deterministic instead of relying completely on generative AI.

These calculations include:

- Urban health scoring
- Urban gap analysis
- Recommendation ranking
- Location suitability
- Growth-demand calculations
- Risk screening
- Feasibility logic

This approach allows the system to provide reproducible and explainable analytical results.

An optional Google Gemini integration can be used to generate explanations and planning summaries from existing analytical results. Gemini is separated from the core numerical decision logic and does not determine database values, calculate authoritative scores, or replace professional planning judgment.

4. FUTURE WORK

Although the current CityMind system provides a functional foundation for urban planning analysis, several improvements can be introduced in future versions.

4.1 Verified Data Integration

Future versions can connect verified demographic, cadastral, transportation, environmental, and infrastructure datasets to improve the accuracy of planning analysis.

4.2 Advanced Spatial Analysis

More advanced GIS capabilities can be introduced using spatial database functionality such as MySQL Spatial or PostGIS.

4.3 Multi-User Collaboration

The platform can be expanded to support different roles such as:

- **Planners**
- **Reviewers**
- **Administrators**
- **Stakeholders**

Users could collaborate on the same planning project with comments, approvals, and revision history.

4.4 Improved Growth Prediction

Historical population and development datasets could be used to build more advanced time-series forecasting models and evaluate prediction accuracy.

4.6 Advanced Recommendation Analysis

The recommendation engine can be extended with:

- Network analysis
- Travel-time calculation
- Multi-objective optimization
- Advanced accessibility analysis

4.7 Mobile Application

A Progressive Web App or native mobile application could allow planners to collect field information directly from mobile devices.

4.8 Advanced Export

Future versions can support:

- PDF
- CSV
- GeoJSON
- GIS project files

4.9 Scalability & Production Infrastructure

Caching, background processing, pagination, monitoring, backups, and stronger production security can be introduced to support larger datasets and more users.

These future improvements are consistent with the current project's documented development direction.

5. CONCLUSION

CityMind was developed as an integrated urban planning and decision-support platform that connects GIS-based planning with structured analytical processes.

The system allows users to define a selected planning area, create blocks and infrastructure, analyze service gaps, calculate urban health indicators, estimate future demand, identify risks, evaluate locations, generate recommendations, assess development feasibility, simulate budgets, and prepare planning reports.

The project combines React 19, Vite, Tailwind CSS, Leaflet, React Leaflet, Recharts, Node.js, Express.js, MySQL 8, REST APIs, JWT, bcrypt, automated testing, and deterministic analytical services into one integrated application. Optional Gemini integration provides natural-language explanations without replacing the core planning calculations.

The project demonstrates how GIS data, structured databases, analytical algorithms, scenario planning, recommendation systems, and modern web technologies can be combined to create a practical urban planning workspace.

The current implementation provides a foundation for further development. Future work can focus on verified real-world datasets, advanced spatial analysis, configurable planning standards, collaboration, scalability, and professional validation before the system is used for formal planning decisions.